

JEPA-style Self-Supervised Learning for Label-Efficient Pneumonia Classification

Datasci 224 Final Project Report

Rebekah Xing

Problem and Dataset

Pneumonia is a leading cause of illness and death worldwide, and it remains one of the leading causes of death in children younger than 5 [7]. Chest radiography, commonly called a chest X-ray, is one of the most frequent initial tests used to detect it [6]. The image reveals areas of opacity in the lung fields that a radiologist interprets as consistent with, or against, infection. Because that interpretation requires trained expert judgment, producing a reliable diagnostic label for each image is slow and costly. This cost is the source of a bottleneck in medical machine learning. Hospitals generate radiographs in very large volumes during regular care, so unlabeled images are abundant, whereas expert-reviewed labels are scarce. Standard supervised learning trains a classifier directly on labeled images and requires many labeled examples to learn good features; when only a few hundred labels are available, it tends to overfit and generalize poorly. Label efficiency can measure a model's ability to achieve high predictive performance using a minimal amount of human-labeled training data.

The research question of this project is whether a model can first learn the structure of chest X-rays from unlabeled images, and then classify pneumonia more accurately than a purely supervised model when only a small fraction of labels is available. The method studied is a Joint Embedding Predictive Architecture (JEPA), a self-supervised approach that learns by hiding part of an image and predicting the latent representation of the hidden region from the visible context, so that useful structure is learned without using any labels [2]. The hypothesis is that this pretraining yields representations that reduce the number of labeled examples needed for accurate downstream classification.

The dataset is PneumoniaMNIST from MedMNIST v2 [1], a binary pediatric chest-X-ray benchmark in which each image is labeled either normal or pneumonia. It is split into 4,708 training, 524 validation, and 624 test images, for a total of 5,856 (Table 1). The classes are imbalanced toward pneumonia in every split because the training set alone contains 1,214 normal and 3,494 pneumonia images, a ratio of approximately 3:1. Although the benchmark is distributed at a default resolution of 28×28 pixels, the images were used here at 224×224 grayscale, because a patch-based self-supervised method requires sufficient resolution for the masked regions to contain meaningful spatial and anatomical structure such as lung fields and opacities. Because of this imbalance, accuracy at a fixed 0.5 threshold can be misleading, since a model that favors the majority class can record a high accuracy while failing to detect the positive cases of clinical interest. The primary metric is the area under the ROC curve (AUROC), which measures how well the model ranks pneumonia cases above normal ones independently of any single threshold.

Joint Embedding Predictive Architecture

The machine learning method used in this project is a Joint Embedding Predictive Architecture (JEPA) [2,3]. JEPA is a self-supervised representation learning method that trains a model using a task created from the data itself instead of expert labels. This project uses a simplified CNN-based JEPA variant rather than the full Vision Transformer version used in I-JEPA [2]. The implementation keeps the core JEPA idea of predicting hidden-region embeddings from visible context, but uses a compact convolutional encoder for the image representation. The main idea behind JEPA is prediction in representation space. Part of an image is hidden, and the model uses the visible part to predict an embedding of the hidden part. Unlike a masked autoencoder [4], JEPA does not try to reconstruct the exact missing pixels. Instead, it

predicts a higher-level representation. This encourages the model to learn structures that may be useful for diagnosis, such as lung fields, opacity patterns, and overall radiographic layout, rather than focusing only on pixel-level texture or noise.

For an input X-ray x , a masking function splits the image into a visible context region x_c and a hidden target region x_t . The visible context is passed through an online context encoder f_θ , producing a context embedding:

$$z_c = f_\theta(x_c)$$

A predictor p_ψ maps this context embedding to a predicted target embedding:

$$\hat{z}_t = p_\psi(z_c)$$

The hidden target region is passed through a target encoder g_ϕ , producing the true target embedding:

$$z_t = g_\phi(x_t)$$

The JEPA pretraining loss is the mean squared error between the predicted target embedding and the true target embedding:

$$L_{JEPA} = \left\| p_\psi(f_\theta(x_c)) - sg(g_\phi(x_t)) \right\|_2^2$$

In this context, $sg(\cdot)$ is the stop-gradient operator. It prevents gradients from flowing into the target encoder during the loss update, so the target embedding acts like a stable regression target. In this implementation, the target encoder is not trained directly by gradient descent. Instead, it slowly tracks the online encoder using an exponential moving average:

$$\phi \leftarrow \tau\phi + (1 - \tau)\theta$$

Where $\tau = 0.996$. The target encoder keeps most of its previous weights and only moves slightly toward the online encoder after each update. This stabilizes training and helps avoid representation collapse, where the model could otherwise learn uninformative constant embeddings [5].

This algorithm is appropriate for the project because the main problem is label efficiency. Chest X-ray labels are expensive, but unlabeled X-ray images are easier to obtain. JEPA fits because it can learn from images without using pneumonia labels. After pretraining, the learned encoder can initialize a classifier trained with only a small fraction of labeled examples. JEPA relies on several assumptions. First, it assumes predictive redundancy. Visible parts of an image should contain useful information about hidden parts. Chest X-rays satisfy this reasonably well because anatomy has consistent spatial structure. Second, it assumes transfer relevance. Features learned from predicting hidden-region embeddings should be useful for pneumonia classification. Third, it assumes stable representation learning. The stop-gradient and EMA target encoder should provide a meaningful learning signal instead of allowing collapse.

Finally, it assumes that enough unlabeled images are available for pretraining. JEPA is expected to perform best when unlabeled data are more available than labeled data, when the input data have strong structure, and when the labeled dataset is too small for a supervised model to learn strong features from scratch. As more labels become available, the advantage of self-supervised pretraining should shrink because the supervised model has enough labeled examples to learn task-specific features directly. This expected pattern is what the label-efficiency experiment is designed to test.

Results and Discussion

The results support the label-efficiency hypothesis. The main comparison was between a CNN trained from scratch and a JEPA-pretrained CNN. JEPA was evaluated in two settings. In the frozen setting, the pretrained encoder was kept fixed and only the classifier head was trained. In the fine-tuned setting, both the encoder and classifier were trained using pneumonia labels. Mean test AUROC across 10 seeds for every setting and label fraction is reported in Table 2 and shown in Figure 1.

Fine-tuned JEPA gives the largest improvement when labels are scarcest. At 1% labels, the scratch CNN reaches 0.801 AUROC, while fine-tuned JEPA reaches 0.910. At 2.5% labels, scratch reaches 0.826 and fine-tuned JEPA reaches 0.919 (Table 2). These results suggest that JEPA pretraining gives the classifier a stronger starting representation when very few labeled examples are available. As the label fraction increases, the gap between scratch training and fine-tuned JEPA becomes smaller. At 5% labels, fine-tuned JEPA reaches 0.924 compared with 0.914 for scratch. At 10% labels, fine-tuned JEPA reaches 0.934 compared with 0.925 for scratch. By 25% labels, scratch reaches 0.931 and fine-tuned JEPA reaches 0.938. This pattern is consistent with the expected behavior of self-supervised pretraining shown in Figure 1. Pretraining helps most when labeled data are scarce, while a supervised model can begin to catch up as more labels become available.

The frozen JEPA results show an important limitation. Frozen JEPA performs well at the lowest label fractions, reaching 0.884 AUROC at 1% labels and 0.887 at 2.5% labels. However, it falls below the scratch CNN at 5%, 10%, and 25% labels (Table 2). This means the pretrained representation is useful, but it is not already a complete pneumonia classifier. Fine-tuning is important because it allows the encoder to adapt its general X-ray features to the specific normal-versus-pneumonia task. The most important observation was that JEPA helped most in the lowest label setting, but fine-tuning was still needed for the representation to become task specific. One limitation is that the experiment uses PneumoniaMNIST instead of a larger clinical dataset, so the results should be interpreted as evidence for label efficiency in a controlled benchmark.

Overall, the results show that JEPA-style self-supervised pretraining improves label efficiency for PneumoniaMNIST pneumonia classification. JEPA does not directly learn pneumonia during pretraining. It learns general chest X-ray structure from unlabeled images. That structure is most helpful when the supervised classifier has very few labels to learn from, and its advantage decreases as more labeled data become available.

References

- [1] Yang, J., Shi, R., Wei, D., Liu, Z., Zhao, L., Ke, B., Pfister, H., and Ni, B. MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10, 41, 2023.
- [2] Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA). *CVPR*, 2023.
- [3] LeCun, Y. A Path Towards Autonomous Machine Intelligence. *OpenReview*, 2022.
- [4] He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. Masked Autoencoders Are Scalable Vision Learners. *CVPR*, 2022.
- [5] Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P. H., et al. Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning. *NeurIPS*, 2020.
- [6] Rajpurkar, P., Irvin, J., Zhu, K., et al. CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning. *arXiv:1711.05225*, 2017.
- [7] McAllister, D. A., Liu, L., Shi, T., Chu, Y., Reed, C., Burrows, J., Adeloye, D., Rudan, I., Black, R. E., Campbell, H., and Nair, H. Global, regional, and national estimates of pneumonia morbidity and mortality in children younger than 5 years between 2000 and 2015: a systematic analysis. *The Lancet Global Health*, 7(1), e47–e57, 2019.

Figures and Tables

Table 1. PneumoniaMNIST class counts. The dataset is imbalanced toward pneumonia in every split.

Split	Normal	Pneumonia	Total
Train	1,214	3,494	4,708
Val	135	389	524
Test	234	390	624

Table 2. Mean test AUROC over 10 seeds with 95% bootstrap CIs. JEPA uses a random masking ratio of 0.6.

Labels	Scratch CNN	Frozen JEPA	Fine-tuned JEPA
1%	0.801 [0.762, 0.835]	0.884 [0.877, 0.891]	0.910 [0.902, 0.917]
2.5%	0.826 [0.798, 0.855]	0.887 [0.879, 0.893]	0.919 [0.915, 0.924]
5%	0.914 [0.904, 0.923]	0.889 [0.883, 0.895]	0.924 [0.916, 0.931]
10%	0.925 [0.916, 0.932]	0.894 [0.889, 0.899]	0.934 [0.926, 0.940]
25%	0.931 [0.928, 0.934]	0.909 [0.905, 0.912]	0.938 [0.931, 0.944]

Figure 1. Test AUROC vs. fraction of training labels for the scratch CNN, frozen JEPA, and fine-tuned JEPA (random masking ratio 0.6). Shaded bands are 95% bootstrap CIs.

